

Tugas Praktik Sistem Kendali 2

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Jawaban

No 1a

```
>> %soal no 1 step respon
```

```
>> % bagian a
```

```
>> num= [5];
```

```
>> den=conv([1 0],[1 5]);
```

```
% s(s+5)
```

```
sys=tf(num, den);
```

```
step(sys)
```

```
>> grid on
```

```
title('Step Response No 1a')
```

```
infoa1= stepinfo(sys)
```

Warning: Simulation did not reach steady state. Please specify YFINAL if this system is stable and eventually settles.

> In [DynamicSystem/stepinfo>localSTEPINFO](#) (line 87)

In [DynamicSystem/stepinfo](#) (line 59)

```
infoa1 =
```

[struct](#) with fields:

RiseTime: NaN

TransientTime: NaN

SettlingTime: NaN

SettlingMin: NaN

SettlingMax: NaN

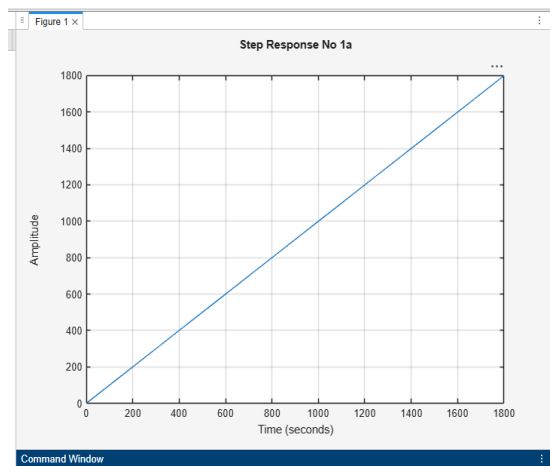
Overshoot: NaN

Undershoot: NaN

Peak: Inf

PeakTime: Inf

```
>> title('Step Response No 1a')
```



No 1b

%soal no 1 step respon

% bagian b

num= [20];

den=conv([1 0],[1 20]);

% s(s+20)

sys=tf(num, den);

step(sys)

grid on

title('Step Response No 1b')

info1b= stepinfo(sys)

info1b =

struct with fields:

RiseTime: NaN

TransientTime: NaN

SettlingTime: NaN

SettlingMin: NaN

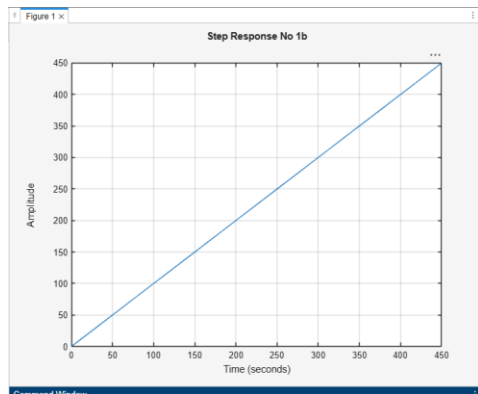
SettlingMax: NaN

Overshoot: NaN

Undershoot: NaN

Peak: Inf

PeakTime: Inf



No 2

```
>> %soal no 2 step respon
```

```
num= [0.703];
```

```
den=[1 0.703];
```

```
sys=tf(num, den);
```

```
step(sys)
```

```
grid on
```

```
title('Step Response No 2')
```

```
info2= stepinfo(sys)
```

```
info2 =
```

struct with fields:

RiseTime: 3.1252

TransientTime: 5.5648

SettlingTime: 5.5648

SettlingMin: 0.9045

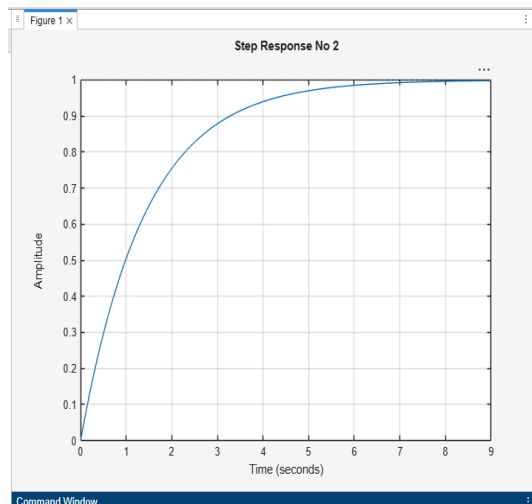
SettlingMax: 0.9993

Overshoot: 0

Undershoot: 0

Peak: 0.9993

PeakTime: 10.415



Soal No 3

```
>> %soal no 3
```

```
% M=1
```

```
sys1=tf([1],[1 6]);
```

```
% M=2
```

```
sys2=tf([0.5],[1 3]);
```

```
step(syss1, sys2)
```

```
legend('M=1' , 'M=2')
```

```
grid on
```

```
title ('step Response no3')
```

Unrecognized function or variable 'syss1'.

Did you mean:

```
>> step(sys1, sys2)
```

```
>> info3_M1 = stepinfo(sys1)
```

```
info3_M2 = stepinfo(sys2)
```

```
info3_M1 =
```

struct with fields:

RiseTime: 0.3662

TransientTime: 0.6520

SettlingTime: 0.6520

SettlingMin: 0.1508

SettlingMax: 0.1666

Overshoot: 0

Undershoot: 0

Peak: 0.1666

PeakTime: 1.2204

info3_M2 =

[struct](#) with fields:

RiseTime: 0.7323

TransientTime: 1.3040

SettlingTime: 1.3040

SettlingMin: 0.1508

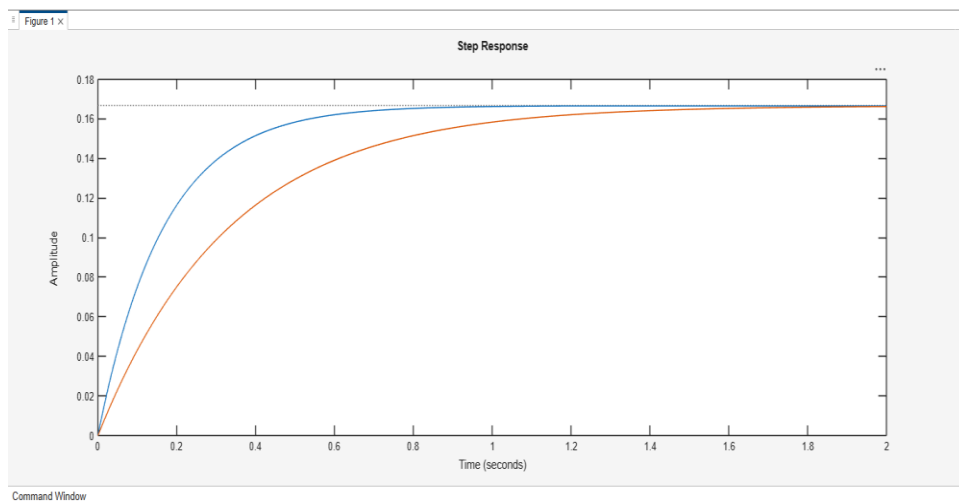
SettlingMax: 0.1666

Overshoot: 0

Undershoot: 0

Peak: 0.1666

PeakTime: 2.4407



Soal no 4a

% Soal no 4a Step Response

```
sys = tf([16],[1 3 16]);
```

```
step(sys)
```

```
grid on
```

```
title('Step Response No 4a')
```

```
info4a = stepinfo(sys)
```

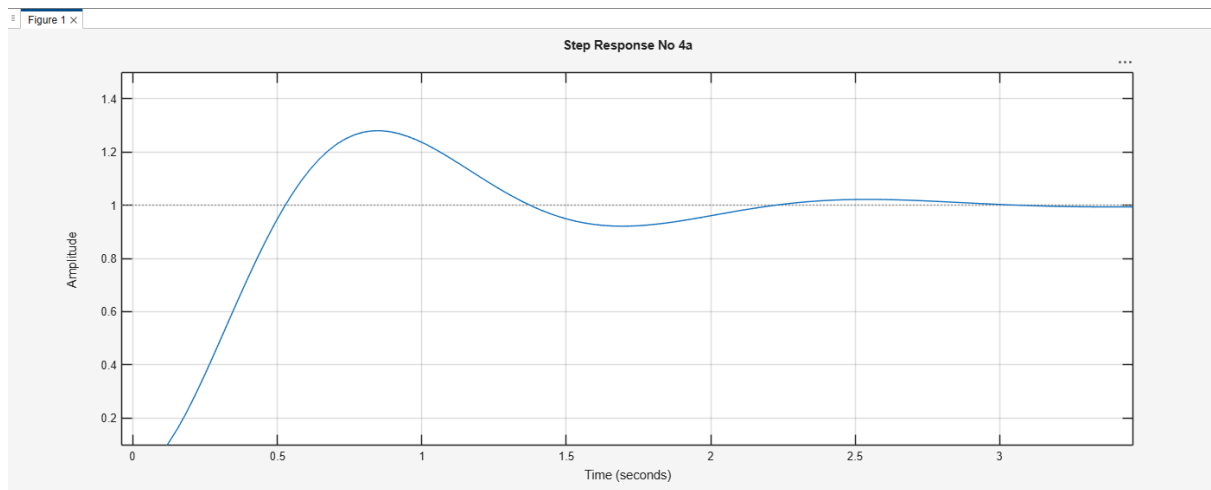
```
info4a =
```

[struct](#) with fields:

RiseTime: 5.4160
TransientTime: 380.0163
SettlingTime: 380.0163
SettlingMin: 0.2699
SettlingMax: 1.8545
Overshoot: 85.4461
Undershoot: 0
Peak: 1.8545
PeakTime: 15.7080
info4a =

[struct](#) with fields:

RiseTime: 0.3569
TransientTime: 2.6570
SettlingTime: 2.6570
SettlingMin: 0.9213
SettlingMax: 1.2803
Overshoot: 28.0255
Undershoot: 0
Peak: 1.2803
PeakTime: 0.8596



Soal No 4b

% Soal no 4b Step Response

```
sys = tf([0.04],[1 0.02 0.04]);
```

```
step(sys)
```

```
grid on
```

```
title('Step Response No 4b')
```

```
info4b = stepinfo(sys)
```

```
s=stepinfo(sys)
```

```
info4b =
```

struct with fields:

RiseTime: 5.4160

TransientTime: 380.0163

SettlingTime: 380.0163

SettlingMin: 0.2699

SettlingMax: 1.8545

Overshoot: 85.4461

Undershoot: 0

Peak: 1.8545

PeakTime: 15.7080

```
s =
```

struct with fields:

RiseTime: 5.4160

TransientTime: 380.0163

SettlingTime: 380.0163

SettlingMin: 0.2699

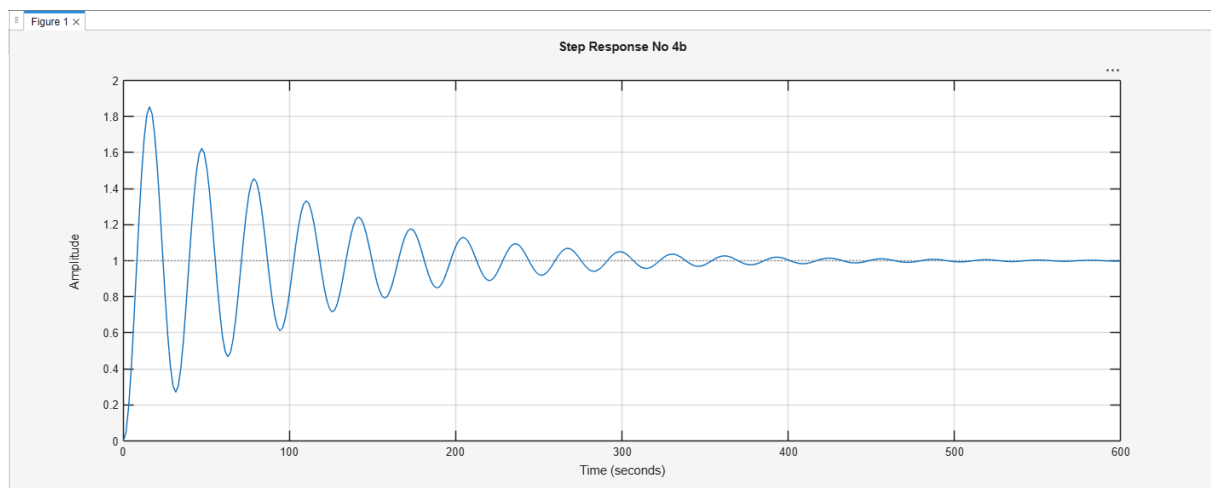
SettlingMax: 1.8545

Overshoot: 85.4461

Undershoot: 0

Peak: 1.8545

PeakTime: 15.7080



Soal No 5a

% Soal no 5

```
>> sys = tf([81.967],[1 40.984]);
```

```
step(sys)
```

```
grid on
```

```
info5a = stepinfo(sys)
```

```
info5a =
```

[struct](#) with fields:

RiseTime: 0.0536

TransientTime: 0.0955

SettlingTime: 0.0955

SettlingMin: 1.8090

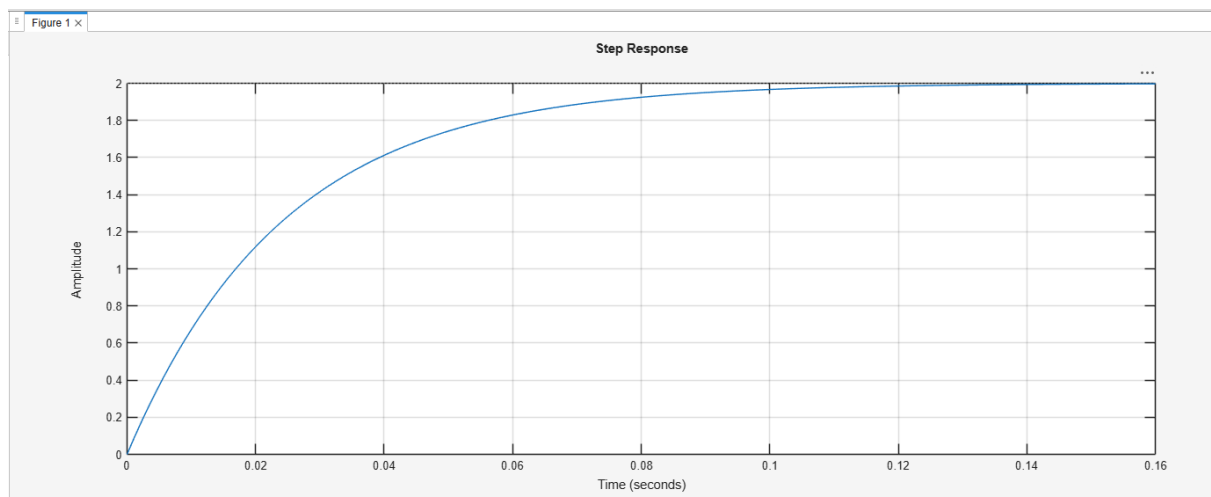
SettlingMax: 1.9987

Overshoot: 0

Undershoot: 0

Peak: 1.9987

PeakTime: 0.1787



Soal No 5b

```
%soal no 5b
```

```
sys = tf([0.669],[1 0.458 0.669]);
```

```
step(sys)
```

```
grid on
```

```
info5b = stepinfo(sys)
```

```
s=stepinfo(sys)
```

```
info5b =
```

[struct](#) with fields:

RiseTime: 1.5895
TransientTime: 16.8853
SettlingTime: 16.8853
SettlingMin: 0.8401
SettlingMax: 1.4000
Overshoot: 39.9971
Undershoot: 0
Peak: 1.4000
PeakTime: 4.0220

